

data lab3

Geekkid

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1)

$$\pi_{firstName,lastName}(\pi_{madeByPid}(\text{Reservation}) \bowtie_{pid=madeByPid} (\text{Person}))$$

2)

$$(\pi_{pid}(\text{attached})) \cup (\pi_{pid}(\sigma_{role=pid})(\text{Participates}))$$

3)

$$\pi_{labId}(\sigma_{building=CollegeofComputing}(\text{Laboratory})) \cup \\ \pi_{labId}(\pi_{modelId}(\sigma_{category=CPUServer}(\text{EquipmentModel})) \bowtie \text{EquipmentUnit})$$

4)

$$\pi_{labId}((\pi_{modelId}(\sigma_{category=CPUServer}(\text{EquipmentModel}))) \bowtie \text{EquipmentUnit}) \cap \\ \pi_{labId}((\pi_{modelId}(\sigma_{category=GPUServer}(\text{EquipmentModel}))) \bowtie \text{EquipmentUnit})$$

5)

$$(\pi_{supervisorPid,labID}(\text{Laboratory})) / (\pi_{labId}(\sigma_{building=CollegeofComputing}(\text{Laboratory})))$$

6)

$$\pi_{pid,projectCode}(\text{Participates}) / \pi_{projectCode}((\pi_{budgetLine}(\sigma_{labId=2}(\text{FundsLab}))) \bowtie \text{FundsPrj})$$

7)

$$\rho(A, \gamma_{madeByPid; COUNT(resId) \rightarrow n}(\text{Reservations})) \\ \rho(C(1 \rightarrow P1, 2 \rightarrow N1, 3 \rightarrow P2, 4 \rightarrow N2), A \times A) \\ (\pi_{P1,P2}(\sigma_{N1 > N2}(C))) \cup (\pi_{madeByPid}(\text{Reservation}) \times (\pi_{pid}(\text{Person}) - \pi_{madeByPid}(\text{Reservation})))$$

8)

$$\rho(A, ((\text{Reserves}) \bowtie (\text{Reservation})) \bowtie (\text{EquipmentUnit})) \\ \rho(B, \gamma_{madeByPid; COUNT(labId) \rightarrow n}(A)) \\ \pi_{madeById}(\sigma_{n > 1}(B))$$

9)

$$\begin{aligned} & \rho(A, \pi_{labId}(\sigma_{name='AI Movement'}(Laboratory))) \\ & \rho(B, \pi_{serialNo}(EquipmentUnit \bowtie A)) \\ & \rho(C, \pi_{resId}(Reserves \bowtie B)) \\ & \pi_{resId}(\sigma_{submissionTS='2026-09'}(Reservation)) \cap C \end{aligned}$$

$$\pi_{resId}(\sigma_{submissionTS='2026-09'}(Reservation) \bowtie Reserves \bowtie EquipmentUnit \bowtie \sigma_{name='AI Movement'}(Laboratory))$$

10)

$$\begin{aligned} & \rho(A, \pi_{pId,projectCode}(\sigma_{role='PI'}(Participates))) \\ & \rho(B, \gamma_{pId,COUNT(projectCode) \rightarrow n}(A)) \\ & \pi_{pId}(\sigma_{n>1}(B)) \end{aligned}$$

11)

$$\begin{aligned} & \rho(A, \pi_{resId,labId}(Reserves \bowtie EquipmentUnit)) \\ & \rho(C, \gamma_{resId}(count(labId) \rightarrow n(A))) \\ & \pi_{approverPid}(\sigma_{n>1}(C) \bowtie Reservation) \end{aligned}$$

12)

$$\pi_{madeByPid}(\sigma_{plannedStart>'6-11-2026' \vee plannedEnd<'6-11-2026'}(Reservation))$$

13)

$$\begin{aligned} & \rho(A, \gamma_{labId,COUNT(serialNO) \rightarrow n}(EquipmentUnit)) \\ & \rho(C, A \times (\gamma_{AVG(n) \rightarrow avg}(A))) \\ & \pi_{labId}(\sigma_{n<avg}(C)) \end{aligned}$$

14)

$$\begin{aligned} & \rho(A, \pi_{labId,approverId,resId}(EquipmentUnit \bowtie Reserves \bowtie (\sigma_{status='approved'}(Reservation)))) \\ & \rho(B, \gamma_{Pid,labId;Count(resId) \rightarrow numApp}(A)) \\ & \rho(C, (\gamma_{labId;MAX(numApp) \rightarrow max}(B))) \\ & \pi_{labId,Pid}(C \bowtie_{max=numApp}(B)) \end{aligned}$$

15)

$$\begin{aligned} & \rho(A, \pi_{MadeByPid,resId}(\sigma_{submissionTS \geq '01-01-2025' \wedge submissionTS \leq '31-12-2025'}(Reservation))) \\ & \rho(B, \gamma_{madeByPid;COUNT(resId) \rightarrow n}(A)) \\ & \pi_{madeByPid}(\sigma_{n \geq 3}(B)) \end{aligned}$$